

FIELD THEORY PART-II

Phase 1: Basic Vector Language and Coordinates

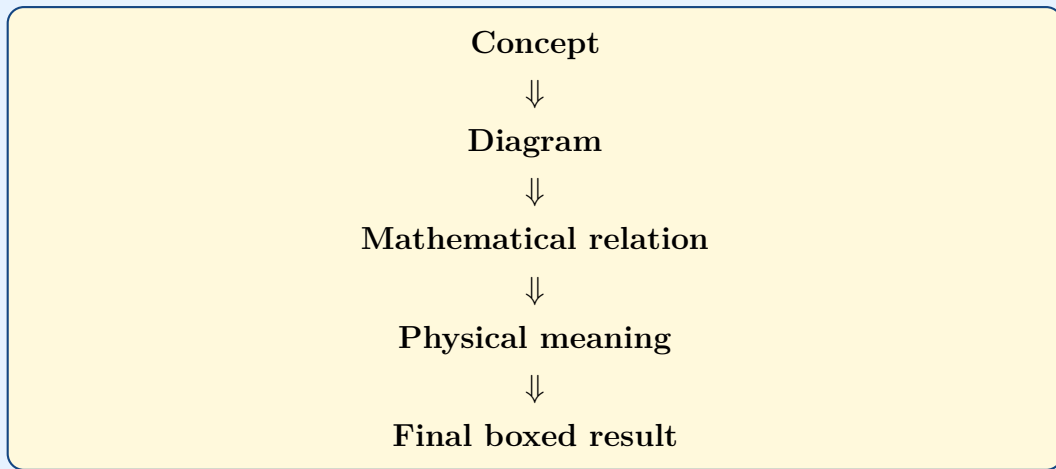
Making the notation friendly before derivations

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Why Phase 1 Is Needed

Main Purpose

Phase 1 builds the language of Field Theory. Before curl, divergence, Stokes' theorem and Maxwell equations, one must understand vector notation and differential elements.

Core Idea

A vector field is written as

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}.$$

Here A_x , A_y and A_z are the components of the vector field along x , y and z directions.

2 Unit Vectors

Definition

A unit vector is a vector having magnitude one. It gives direction only.

Cartesian Unit Vectors

In Cartesian coordinate system, the three basic unit vectors are:

$$\hat{\mathbf{x}}, \quad \hat{\mathbf{y}}, \quad \hat{\mathbf{z}}.$$

They are mutually perpendicular:

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = 0, \quad \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = 0, \quad \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0.$$

Also:

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = 1, \quad \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = 1, \quad \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1.$$

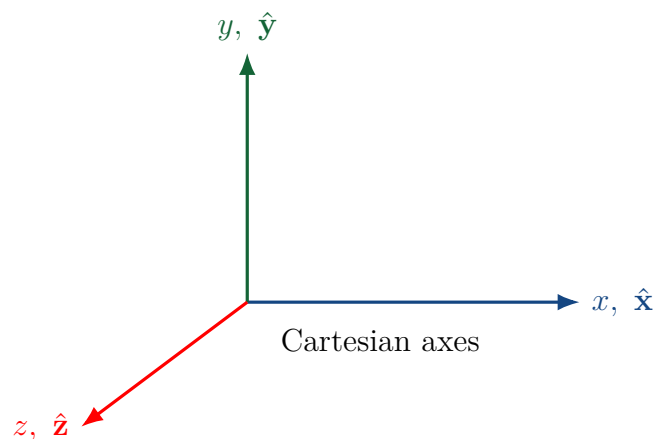


Figure 1: Three mutually perpendicular unit vectors in Cartesian coordinate system.

3 Vector Field

Meaning

A vector field gives a vector at every point in space.

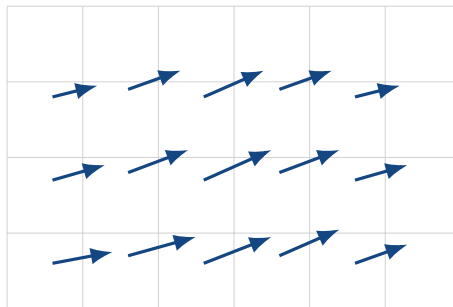
Example

Electric field is a vector field:

$$\mathbf{E} = E_x \hat{\mathbf{x}} + E_y \hat{\mathbf{y}} + E_z \hat{\mathbf{z}}.$$

Magnetic field intensity is also a vector field:

$$\mathbf{H} = H_x \hat{\mathbf{x}} + H_y \hat{\mathbf{y}} + H_z \hat{\mathbf{z}}.$$



A vector field has direction and magnitude at many points.

Figure 2: Simple representation of a vector field.

4 Differential Length

Definition

A differential length vector is a very small directed length element.

Cartesian Form

In Cartesian coordinates,

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}.$$

For a small length along x only:

$$d\mathbf{l} = dx \hat{\mathbf{x}}.$$

For a small length along y only:

$$d\mathbf{l} = dy \hat{\mathbf{y}}.$$

For a small length along z only:

$$d\mathbf{l} = dz \hat{\mathbf{z}}.$$

Why It Matters

Line integral expressions such as

$$\oint_C \mathbf{H} \cdot d\mathbf{l}$$

use differential length. This expression means adding the component of $\mathbf{H}$ along the path.

5 Differential Surface

Definition

A differential surface vector is a small area with a direction normal to the surface.

Common Surface Elements

For a surface perpendicular to x :

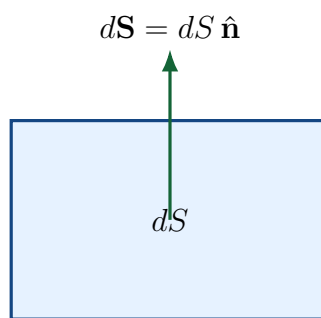
$$d\mathbf{S} = dy \, dz \, \hat{\mathbf{x}}.$$

For a surface perpendicular to y :

$$d\mathbf{S} = dx \, dz \, \hat{\mathbf{y}}.$$

For a surface perpendicular to z :

$$d\mathbf{S} = dx \, dy \, \hat{\mathbf{z}}.$$



Surface element with outward normal

Figure 3: Differential surface vector.

6 Differential Volume

Small Box

A small rectangular volume is:

$$dV = dx \, dy \, dz.$$

Divergence derivation uses such a small volume.

7 Line Integral

Meaning

A line integral adds the component of a vector field along a path.

Closed Line Integral

The expression

$$\oint_C \mathbf{A} \cdot d\mathbf{l}$$

means the circulation of $\mathbf{A}$ around a closed curve C .

8 Surface Integral

Meaning

A surface integral adds the component of a vector field normal to a surface.

Flux Integral

The expression

$$\iint_S \mathbf{A} \cdot d\mathbf{S}$$

means flux of $\mathbf{A}$ through surface S .

9 Exam Connection

What Phase 1 Unlocks

After Phase 1, these expressions should look familiar:

$$\oint_C \mathbf{H} \cdot d\mathbf{l}, \quad \iiint_S \mathbf{J} \cdot d\mathbf{S}, \quad \iint_S \mathbf{B} \cdot d\mathbf{S}.$$

These are the language of Ampere's law, Biot-Savart law, Stokes' theorem and divergence theorem.

10 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.